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Monsoon, 2024

B. Tech 3<sup>rd</sup> Semester Examination

Building Planning and Computer – Aided Civil Engineering Drawing

Course Code: BCE23303T

Full Marks – 50

Time – 2 hours

*The figure in the margin indicates full marks for the questions.*

(10x1)

Q1. Answer the following questions

- (i) One- or Two-Family Private Dwellings falls under which group of building
  - a. G-A1
  - b. G-A2
  - c. G-A3
  - d. G-A4
- (ii) Low-emitting materials credit applies to different products like:
  - a. Interior paints and coatings
  - b. Thermal and acoustic insulation
  - c. Both a. and b.
  - e. Neither a. nor b.
- (iii) The National Building Code of India (NBC), was first published in the year
  - a. 1970
  - b. 1983
  - c. 1987
  - d. 2005
- (iv) What will be the maximum allowable percentage of built-up area of a plot having 1000sqm
  - a. 60%
  - b. 50%
  - c. 40%
  - d. 66%

- (v) Ocean water is unfit for drinking and industrial or agricultural uses as it contains about ..... grams per liter of dissolve minerals or salts.
- 40
  - 70
  - 88
  - 35
- (vi) Which color of the bin refers to the sanitary waste collection.
- Green
  - Blue
  - Yellow
  - Red
- (vii) Cleat wiring can only be used on ..... Volts on low temperature
- 250/440
  - 250/420
  - 250/430
  - 250/450
- (viii) The pitch of the stair must not exceed .....and minimum of .....
- $50^\circ$  and  $35^\circ$
  - $40^\circ$  and  $25^\circ$
  - $40^\circ$  and  $25^\circ$
  - $40^\circ$  and  $25^\circ$
- (ix) Geared Traction Elevator can get a movement speed of up to ..... feet per minute. The peak movement distance provided by geared traction lifts is ..... feet.
- 300 and 600
  - 500 and 250
  - 300 and 500
  - 500 and 2000
- (x) An escape ladder provided in fire resisting building should be at least ..... wide.
- 100cm
  - 90cm
  - 150cm
  - 200cm

Q2. a. Explain the various types of building outlined by National Building Code of India. Discuss the principles of building planning and explain the factors affecting them in brief. (10)

OR

b. Write the names of low emitting building materials. Also Explain in brief their application in a commercial building. Write 10 aspects covered by the building bye-laws and explain each in brief (10)

Q3. a. A plot owner wants to construct a building having 4 floors over the plot area of 4000 sq ft. The FAR of the locality is 2.5. What will be the covered area of the building, if all the floors are constructed having the same area? (10)

OR

b. Calculate the FAR of a building having 3 floors built over 1200 sq ft. area. The area of each floor is 650 sq ft. Also calculate the total floor area of the building which you can construct over the plot of size 50ft.x 80ft. The permitted FAR of that particular city is 2. (10)

Q4. a. Elaborate the steps involved in purification of water before supplying in details. Conclude the importance of distribution system in brief. (10)

OR

b. Define Waste segregation. Differentiate wet waste and segregation, also signify the Significance of Waste Segregation. (10)

Q5. a. Create the figure to show the components of staircase and explain each component in brief. (5)

b. Discuss the different types of staircases and mention the requirements of ideal staircase along with the dimensions. (5)

OR

c. Create the plan of a stair hall of an educational building, which measures 5.25mX6.25m. The vertical distance between the floors is 4.2m. Design a suitable stair for the building. (10)